OMIS106 Final Project:

Brianna Park, Wei Wang, Sabrina Chu, Asher Dahlke, Benjamin Haak, Chatham Griffiths, Divya Bengali, and Ryan Lo

Part 1:
Step 1.1: Define the app based on **one of these three** possible trigger points:

- Respond to an opportunity
- **Resolve a problem**
- Respond to an external directive

Step 1.2: Generate System Vision Document, which includes:

- Problem Description (Paragraph)

Many communities lack a centralized service platform to efficiently connect volunteers to organizations in need of their services. Existing platforms struggle with decentralization which leads to fragmented, inefficient approaches to volunteer selection. This inefficiency results in increased difficulty for organizations who struggle to recruit and manage volunteers as well as volunteers who struggle to find opportunities that fit their interests and schedule. This project aims to create a centralized platform which can aid organizations in volunteer communication, management, and scheduling. The purpose of this project is to streamline volunteer recruitment, management and scheduling with the ultimate goal of increasing participation rates and improving the impact of volunteer initiatives.

- System Capabilities (Bullet)
    - Multi-Criteria Opportunity Search & Filtering
    - Smart One-Click Applications with Automatic Qualification Check
    - Organization Opportunity Posting
    - Applicant Review, Decision, and Notification Dashboard
    - Two-Way Offer Management with Integrated Scheduling
    - In-App Messaging
    - Personalized Opportunity Recommendations with Learning Engine
    - Two-Way Rating & Review System Post-Event
    - Verified Hours Tracking with Compliance Reporting
    - Subscription Management with Plan-Based Feature Access
- Business Benefits (Bullet)
    - The organization pays our platform
        - Free unlimited listing for a month (so orgs can try it out)
        - Basic Subscription -> unlimited listings, basic analytics
        - Premium and more costly subscription for companies to access -> featured listings, more detailed analytics
    - Free for users
        - Premium subscription for app to track organizations, hours, reviews, priority view (orgs will see you first when applying for volunteer position even if other applied first)
    - Opportunity to find volunteer opportunities easily

        - Rather than having to find individual opportunities, all verified opportunities show in one screen to increase the chance of a volunteer signing up.
    - One application for all postings
        - Since the user only has to provide verification and list skills once in profiles, they don't need to fill out a new application for every listing which will increase the chances of them applying.
    - Personalized opportunities through AI allow for greater listing visibility

---

Part 2:

Step 2.1: Generate User Stories (minimal 10)

- As a volunteer, I want to filter opportunities by organization type, location, date range, required qualifications to find roles that match my skills.
  Acceptance: can combine filters; see counts before applying; saving a search creates an alert.
- As a volunteer, I want to apply to an opportunity based on my profile, containing skills/resume and a recent background check status, so that I can submit complete applications in one click.
  Acceptance: shows required quals vs. mine; blocks submit if required background check is missing; tracks application status.
- As a volunteer, I want to receive one background check for my profiler so that organizations do not need to run an additional background check each time.
- As an organization admin, I want to create an org profile and post opportunities with role description, qualifications, and screening questions, so that applicants understand expectations.
  Acceptance: cannot publish post without required fields; preview before publish; duplicate/edit draft.
- As an organization admin, I want to view applicants' profiles, qualifications, and background-check status and then accept/reject or waitlist with a template message, so that I can staff roles efficiently.
  Acceptance: bulk actions; audit trail of decisions; auto-notify applicants.
- As a **volunteer**, I want to accept or decline an offer that is based on my availability, so that I can confirm shifts without email back-and-forth.
  Acceptance: offer expires after N days; conflict warnings; confirmed shift appears on both sides when accepted.
- As an organization admin, I want to schedule volunteers for an existing volunteer listing in an easy and visible way to ensure all shifts are filled.
- As a volunteer or org admin, I want an in-app message thread tied to an opportunity, so that all communications stay organized and searchable without having to provide

personal contact information.
Acceptance: templates for common messages; attachment support; read receipts.

- As a volunteer, I want the app to recommend opportunities based on my past applications, saved searches, skills, and reviews, so that I discover better-fit roles faster.
  Acceptance: explain-why chips (e.g., “matched your CPR cert”); hide/never-show toggle; improves after feedback.
- As a volunteer, I want to rate and review organizations after a completed shift (and vice versa), so that future matches are higher quality and issues are visible.
  Acceptance: review prompts only after verified completion; report-abuse flow; aggregate scores on profiles.
- As a volunteer, I want to view reviews and comments of a profiler so that I can request to volunteer in a good environment.
- As a **volunteer completing school/court/ employer-mandated hours**, I want to track verified hours against a target and export a signed report, so that I can prove compliance.
  Acceptance: target hours set per directive; supervisor e-signature; downloadable PDF summary.
- As a **platform admin**, I want organizations to choose a subscription plan (limits, premium filters, featured posts) and pay securely, so that we monetize sustainably.
  Acceptance: proration on plan change; invoice history; posting/feature gates tied to plan.

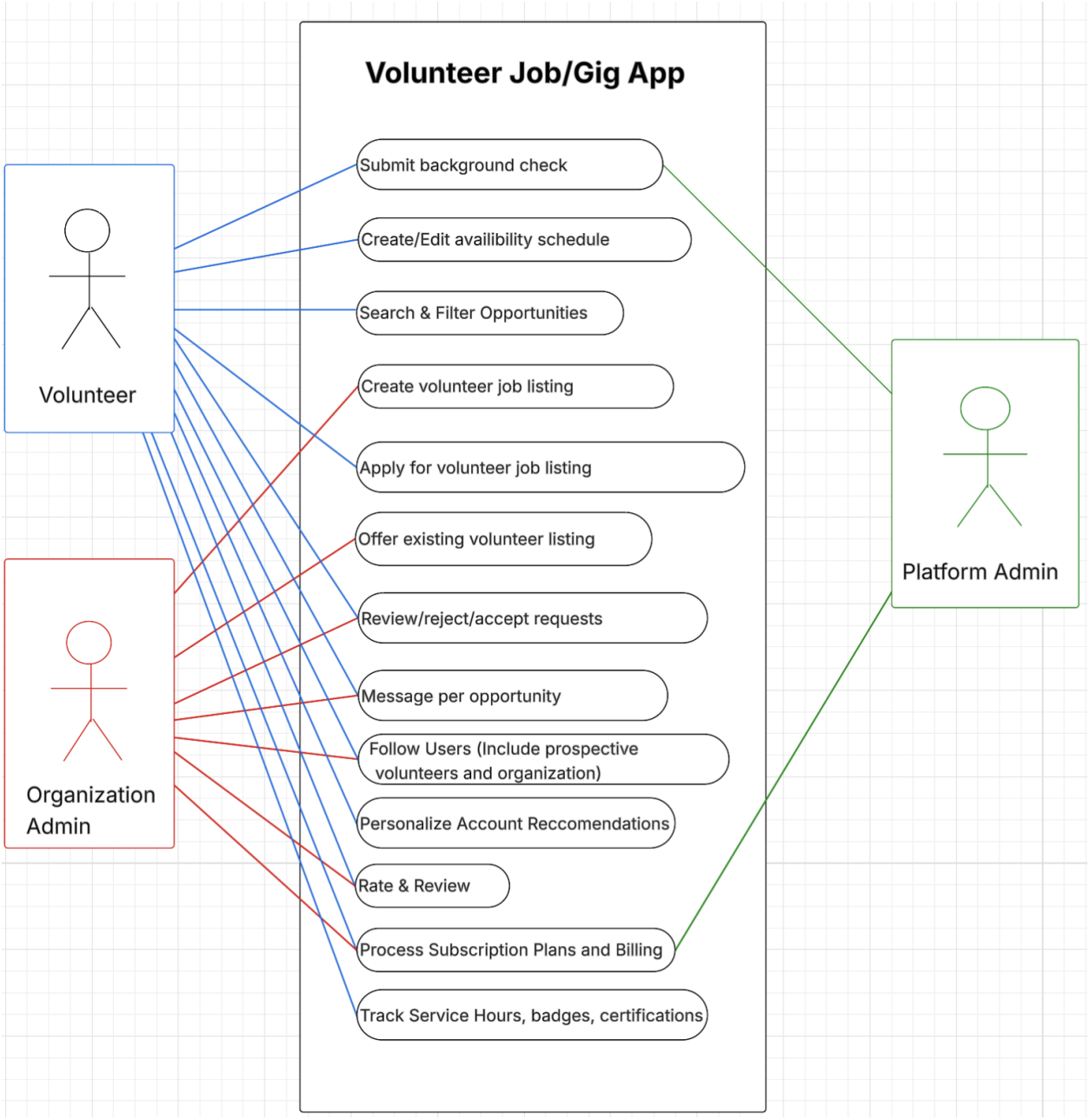
Step 2.2: Based on the user story, generate Use Cases (minimal 10)

1. **Submit background check (finger print scan and clean/criminal record)**
   - Volunteers upload a basic background check with a finger print scan and official record. System records the upload and admin verify the information within 3-5 business days. Admin sends a notification message to volunteer based on if the submission was adequate.
2. **Create/Edit availability schedule**
   - Volunteers enter general availability. System displays a weekly schedule (Mon-Sun) and volunteer enters time slots days/times available. System updates the schedule and maintains it for the volunteer.
3. **Search & Filter Opportunities**
   - Volunteer uses search engine and filters search for volunteer listings based on location, organization type, date/time, and duration. System returns applicable listings based on filtering and search entered.
4. **Create volunteer job listing**
   - Organization creates volunteer listing by entering all pertinent information (title of job, description, date, start/end time, location,, skills, current status, listing expiration date, and number of volunteers needed). When organization publishes listing, system displays it on organization’s profile, records the date of listing, and makes it accessible through the search engine.
5. **Apply for volunteer job listing**

- Volunteer requests to do a volunteer listing for an org. System sends a notification to the organization of volunteer's request, maintains request is in pending state, and adds the volunteer's account name to the list for that listing.

6. **Offer existing volunteer listing opportunity**
    - Organizations offer existing volunteer job listings to volunteers. System send a notification to the volunteer to accept or decline volunteer job offer. System maintains that the offer is in pending status.
7. **Review/reject/accept requests for volunteer jobs**
    - Organizations and volunteers accept or deny pending volunteer job requests from the list of volunteers. System links the name of each volunteer on the list to their profile for quick viewing. Organizations can accept or deny and the system sends a notification to the sender upon decision. If accepted, volunteer will be moved to the accepted volunteer list and the volunteer job listing will be updated to reflect the amount of availability left for the listing.
8. **Message Thread per Opportunity**
9. **Follow users**
    - A user follows another user. System updates follower number and adds list for the user, If followers are restricted, the system sends an accept or deny request to the receiver.
10. **Personalize Account and Recommendations**
    - Volunteers personalize volunteer job listing recommendations based on information selected. System displays options regarding: organization type, social issues, location, skills, location, etc. Based on volunteer's selected items, system recommends job listings, organizations to follow, and related notifications.
11. **Rate and review**
    - Users rate organization or volunteer from a 1 to 5 star rating system and can enter a written review after a volunteer event has occurred. System updates user's average rating after a rate/review is entered and notifies user of written review if user writes one.
12. **Process Subscription plans & Billing**
    - User selects a subscription plan based on the available options. System displays payment options and user inputs desired payment option with billing information. System updates account settings and tools to the subscription plan selected and maintains that billing information is secure.
13. **Track service hours, badges, Certifications**
    - With a premium subscription, volunteers track their service hours and display certifications/badges earned from completing volunteer jobs. System tracks all volunteer hours completed and provides analytics based on those hours. System displays any certifications or badges earned by volunteers on the volunteer profile.

Step 2.3: Generate Use Case Diagrams (minimal 1)

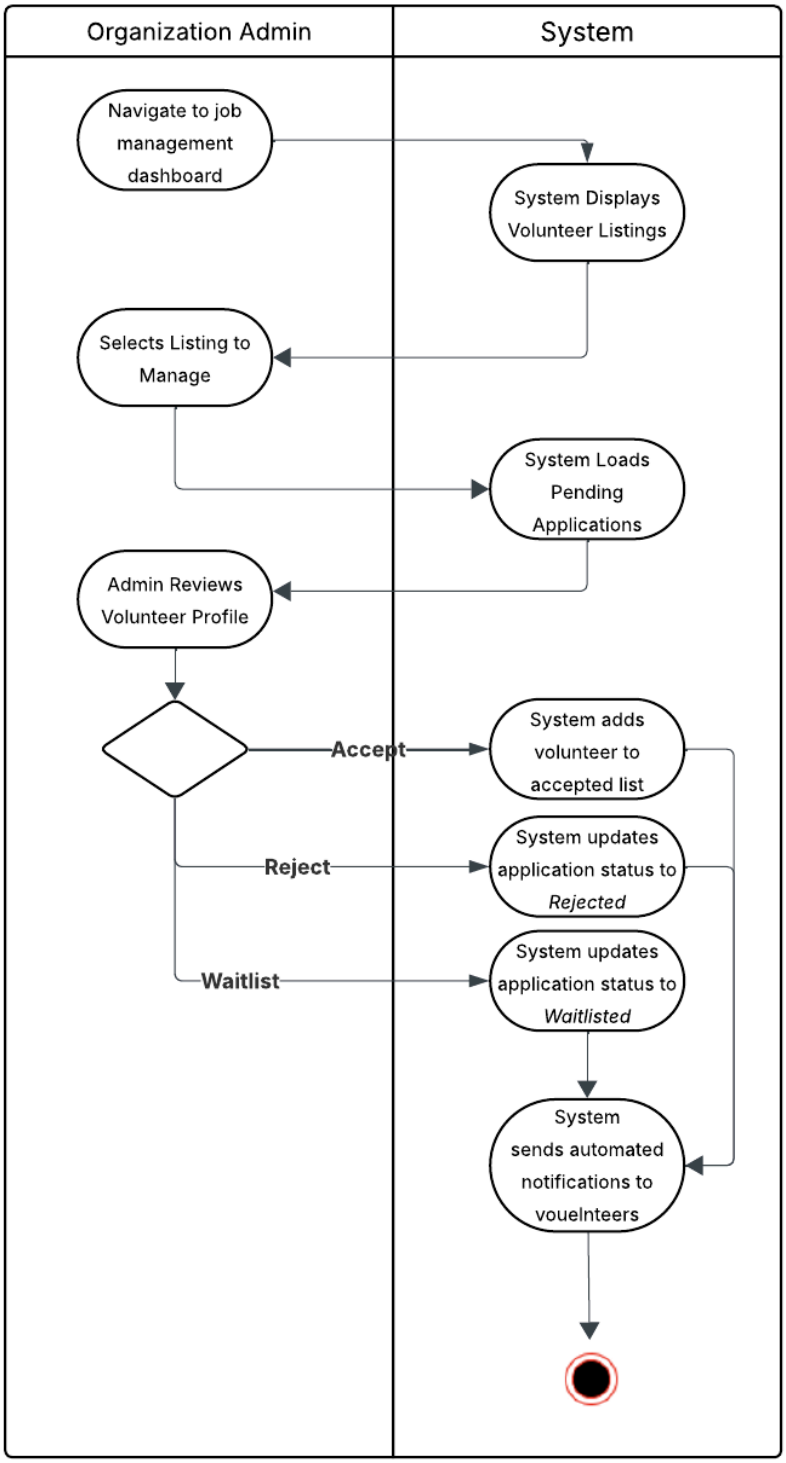


Step 2.4: Generate Activity Diagrams (minimal 1) and fully developed use case descriptions (minimal 1) Asher

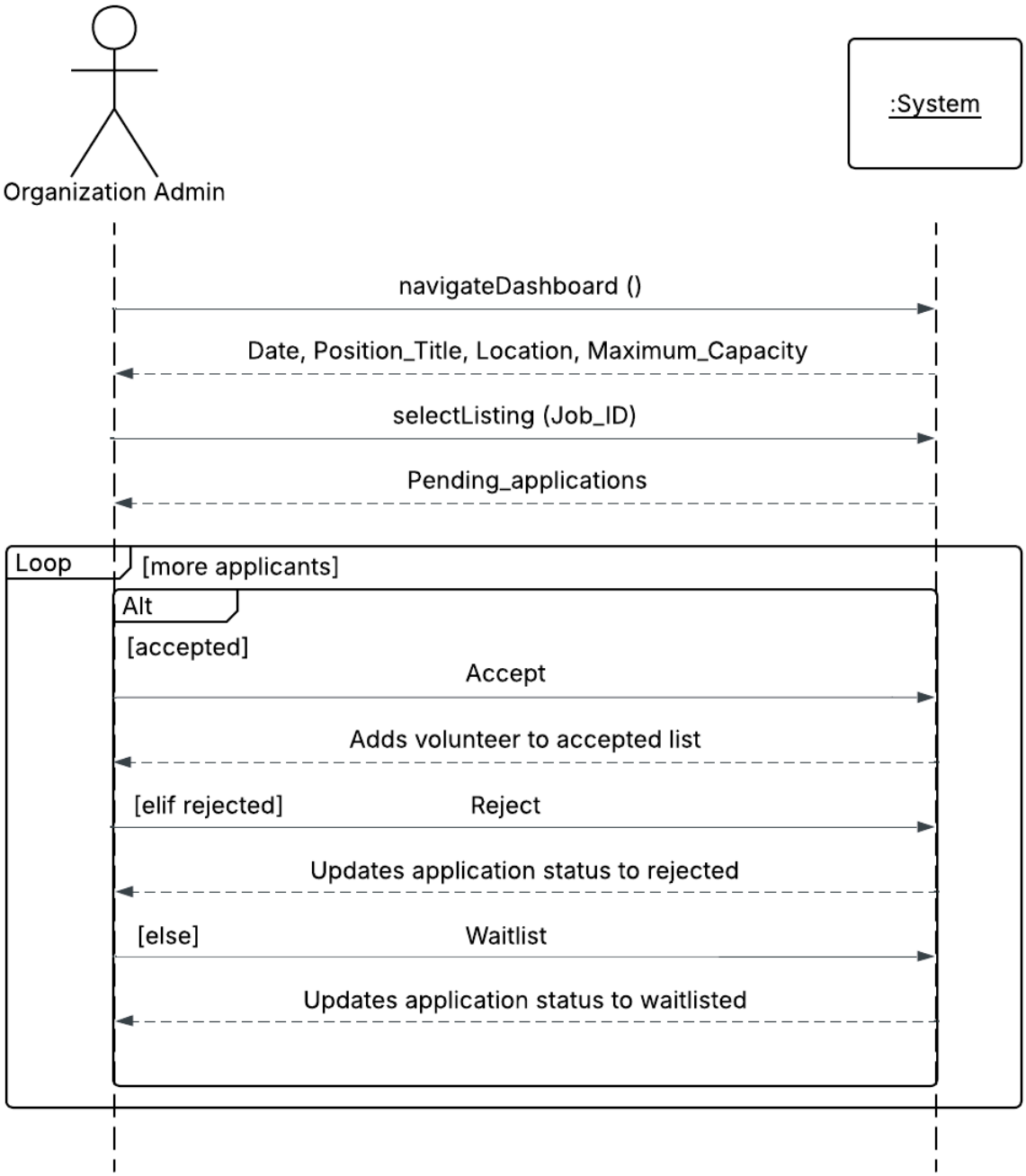
Fully developed use case description and activity diagram

Use case name	**Review/Reject/Accept Requests for Volunteer Job**
Scenario	Organization reviews pending volunteer applications and decides whether to accept, reject, or waitlist them for an existing volunteer job listing.
Triggering event	Organization administrator selects a volunteer job listing with pending applications and chooses to review requests.
Brief description	The organization administrator reviews volunteer applications submitted for an open volunteer job listing. The system displays all pending requests, allows the administrator to view volunteer profiles, and record a decision (accept, reject, or waitlist). Once the decision is made, the system updates the volunteer's application status, sends an automatic notification to the volunteer, and updates the available positions in the listing.
Actors	Organization Administrator
Related use cases	Apply for Volunteer Job Listing Offer Existing Volunteer Listing Opportunity Track Volunteer Hours
Stakeholders	**Organization Administrator, Volunteer, System Administrator**
Preconditions	The organization has an active account and at least one posted volunteer job listing. One or more volunteer applications exist in a *Pending* state. The organization administrator is logged in to the platform.
Post conditions	The volunteer's application status is updated to *Accepted*, *Rejected*, or *Waitlisted*. The system sends a notification to the volunteer informing them of the decision.

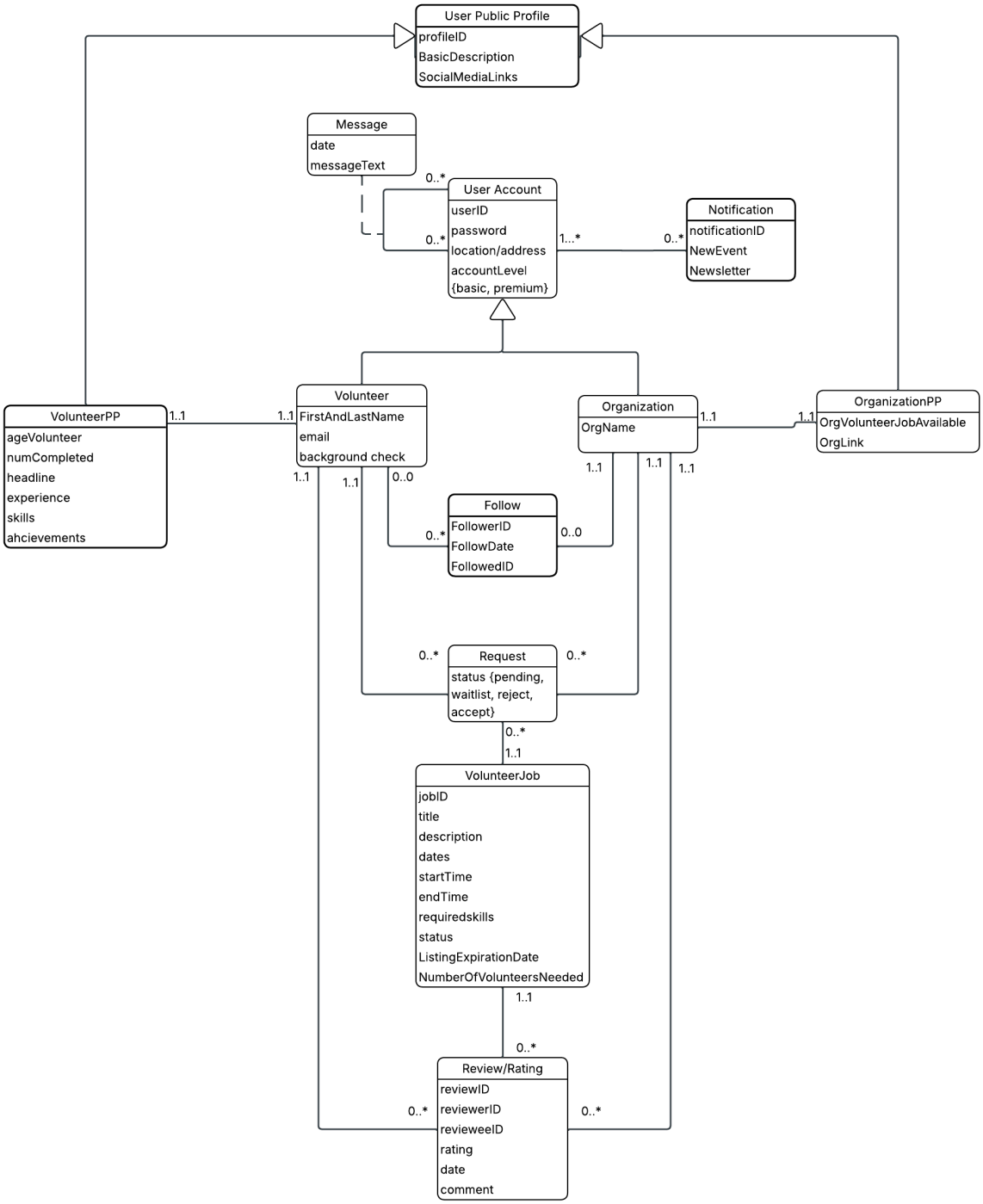
	If accepted, the volunteer is added to the confirmed volunteer list, and the job listing's available slots are reduced.	
Flow of activities	Actor	System
	1. Organization admin navigates to the job management dashboard. 2. Admin selects a specific job listing to review applications. 3. Admin clicks a volunteer's name to view profile details (skills, certifications, background check, reviews). 4. Admin chooses to **Accept**, **Reject**, or **Waitlist** the volunteer. 5. If **Accepted**, admin confirms slot assignment. 6. If **Rejected**, admin confirms decision. 7. If **Waitlisted**, admin confirms waitlist action. 8. Admin reviews updated status list.	1.1 System displays all active volunteer listings with pending applications. 2.1 System retrieves and displays pending volunteer requests for that listing. 3.1 System presents volunteer profile data and related documents. 4.1 System records decision, updates application status, and timestamps the action. 5.1 System adds volunteer to accepted list and decrements available positions. 6.1 System archives the application under rejected list. 7.1 System marks application as "Waitlisted" and retains it for future selection. 8.1 System refreshes dashboard and displays updated status for all applications. 8.2 System sends automatic notification (in-app and email) to each volunteer reflecting decision.
Exception conditions	1.1 No pending applications available. 2.1 Volunteer withdraws before decision. 3.1 Job listing reaches full capacity before acceptance. 4.1 Network or system error during update.	


Organization Admin
System
Navigate to job management dashboard
System Displays Volunteer Listings
Selects Listing to Manage
System Loads Pending Applications
Admin Reviews Volunteer Profile
Accept
System adds volunteer to accepted list
Reject
System updates application status to *Rejected*
Waitlist
System updates application status to *Waitlisted*
System sends automated notifications to vouelnteers

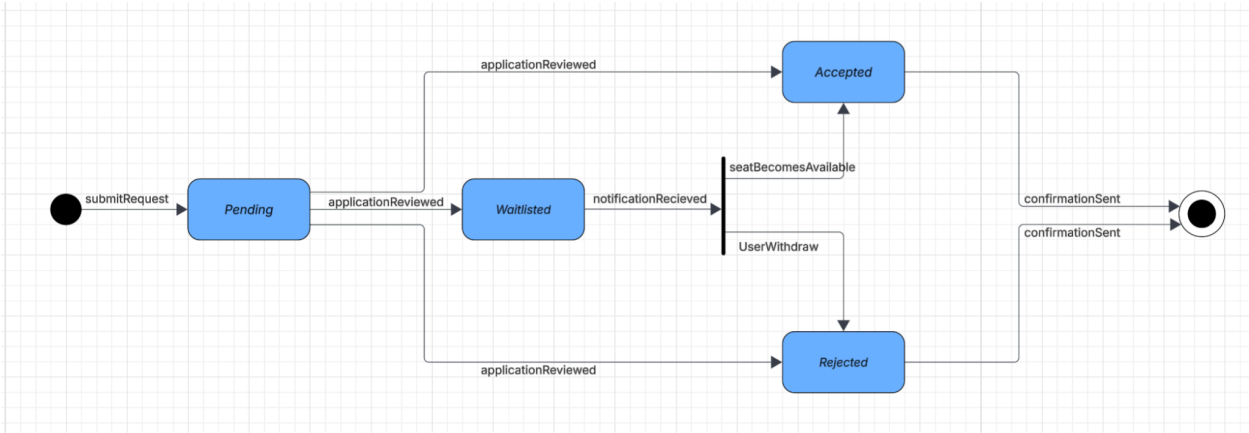
Step 2.5: Generate System Sequence Diagrams (SSD)



Step 2.6: Generate Domain Model Class Diagrams (minimal 1)



Step 2.7: Generate State Machine Diagrams



This state machine diagram refers to the request domain class from the domain class diagram which is between the volunteer and organization.

Part 3: (based on 1.2)
Step 3.1: Build a Cost/Benefit Analysis chart (and state your assumptions)—> NPV, ROI, payback, IRR

Plan	Monthly Price	Assumed % of Users
Free	$0	40%
Basic	$25/mo ($300/yr)	40%
Premium	$45/mo ($540/yr)	20%

Year	Total Orgs	Paying Orgs (60%)	Est. Basic (40%)	Est. Premium (20%)
Y1	20	12	8	4
Y2	50	30	20	10
Y3	100	60	40	20
Y4	150	90	60	30
Y5	200	120	80	40

Year	Basic Revenue (x $300)	Premium Revenue (x $540)	Total Revenue
Y1	$2,400	$2,160	$4,560
Y2	$6,000	$5,400	$11,400
Y3	$12,000	$10,800	$22,800
Y4	$18,000	$16,200	$34,200
Y5	$24,000	$21,600	$45,600

*Development Cost of 20,000 dollar is based of industry average for small startup developing platform.

Year	0	1	2	3	4	5	5 yr Total
Value of Benefit	$0.00	$4,560.00	$11,400.00	$22,800.00	$34,200.00	$45,600.00	
Developmental Cost	-$20,000.00	-$5,000.00	-$3,000.00	-$2,000.00	-$2,000.00	-$2,000.00	
Other expenses	-$3,000.00	-$2,000.00	-$2,000.00	-$2,000.00	-$2,000.00	-$2,000.00	
Net Benefit/Cost	-$23,000.00	-$2,440.00	$6,400.00	$18,800.00	$30,200.00	$41,600.00	$94,560.00
Discount Factor 10%	1.000	0.909	0.826	0.751	0.683	0.621	
NPV	-$23,000.00	-$2,218.18	$5,289.26	$14,124.72	$20,627.01	$25,830.33	$63,653.13
Cumulative NPV	-$23,000.00	-$25,218.18	-$19,928.93	-$5,804.21	$14,822.80	$40,653.13	
Payback Period	3.281388742	year					
ROI with Discount	276.75%						
ROI without Discount	411.13%						
IRR	41%						

Intangible Benefits:

Being a volunteer service platform, we believe that our product provides a benefit to the communities we serve beyond a strict monetary value. We care about community welfare and believe that connecting volunteers to service organizations allows them to most efficiently share their passions and skills with where they can be most impactful. A strong part of the value we generate will be represented in the strength of our communities, the fulfilment of our volunteers, and the impact of our organizations:

- **Sense of purpose and community engagement**: Connect individuals with meaningful causes, fostering a deeper sense of belonging and shared responsibility.
- **Building social capital and networking**: Volunteers and organizations can expand their networks, creating valuable personal and professional connections.
- **Skill development and personal growth**: Volunteering offers opportunities to gain new skills, enhance experience, and foster personal growth for everyone involved.
- **Flexibility and reduced barriers to participation**: Offers flexible opportunities, allowing volunteers to contribute according to their availability and interests, increasing participation.
- **Emotional rewards and fulfillment**: Both volunteers and organizations experience emotional satisfaction, with volunteers feeling accomplished and organizations benefiting from engaged, motivated supporters.
- **Increased engagement with causes**: Individuals are connected to causes they care about, promoting sustained involvement and a deeper commitment to social impact.
- **Social impact and ripple effect**: Volunteers inspire others to get involved, amplifying the overall impact of the organization's efforts and encouraging a culture of giving back.
- **Increased awareness and advocacy**: Help spread awareness of causes, with volunteers becoming passionate advocates who further elevate the visibility and reach of organizations' missions.

Step 3.2: Build a Project iteration schedule with a detailed task breakdown (need to consider dependencies among tasks)

Iteration	Time Estimate	Use Case Assigned to Iteration
1	2 weeks	1. Account Creation a. Create and manage account b. Submit background check c. Create availability schedule d. Add/store personal information e. Email verification f. Password Reset
2	2 weeks	1. Create a volunteer job listing 2. Search and filter opportunities 3. Save/favorite opportunities 4. Apply for a volunteer job listing 5. Review requests for volunteer jobs 6. Send notifications when accepted or rejected
3	1.5 weeks	1. Message thread per opportunity 2. Follow organizations 3. Feed for organizations 4. Notification Settings 5. Rate and Review
4	1.5 weeks	1. Process subscription plans and billing 2. Organization exclusive dashboard a. Track/submit volunteer hours, volunteers, and listings 3. Organization confirms volunteer hours 4. Dashboard for personal service hours 5. Export hours and participation 6. Admin management for plans
5	1 week	1. Personalize account and recommendations 2. Performance monitoring (for errors or site breaks) 3. Security checks 4. Clean up mock-up, final test, harden site
Total	8 weeks	

Step 3.3: Build a Work Breakdown Structure
Iteration 1 – Account Creation Subsystem**

I. Project Planning

- Develop WBS and schedule – ½ day
- Team kickoff meeting – ½ day

II. Analysis Tasks

- Meet with stakeholders on account-related requirements – 1 day
- Define data needs for account, background check, and schedule – 1 day
- Analyze user workflow for account setup – 1 day
- Model user activities – 1 day

III. Design Tasks

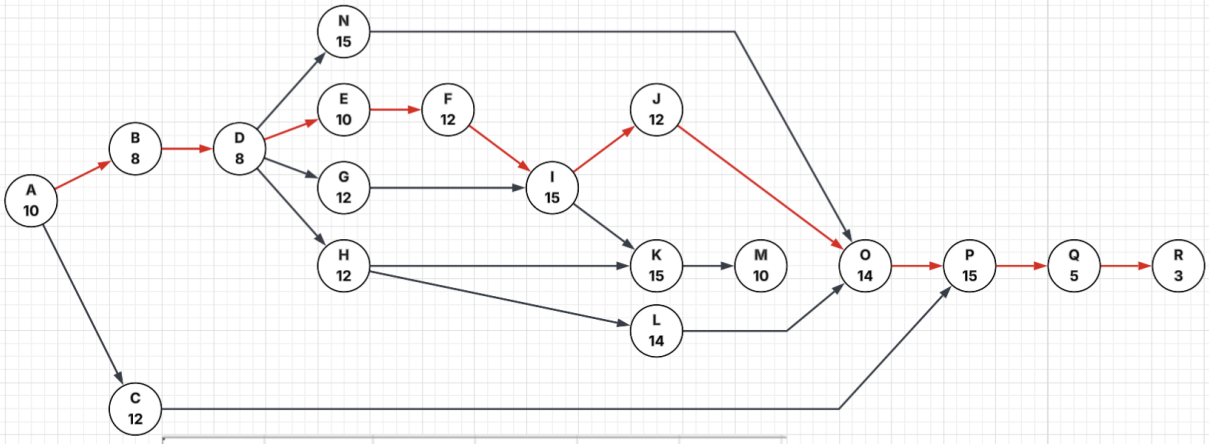
- Design database schema – 1 day
- Create UI wireframes for account creation and related screens – 2 days
- Identify program classes/methods – 1 day

IV. Build Tasks *(shorter & brief)*

- Build database tables – ½ day
- Implement core account creation & management features – 2 days
- Develop background check submission – 1 day
- Develop availability schedule setup – 1 day
- Add email verification & password reset – 1 day
- Set up test data – ½ day
- Perform acceptance tests – 1½ days
- Release version & team retrospective – 1 day

Step 3.4: Build a Project Network Diagram —> PERT chart?

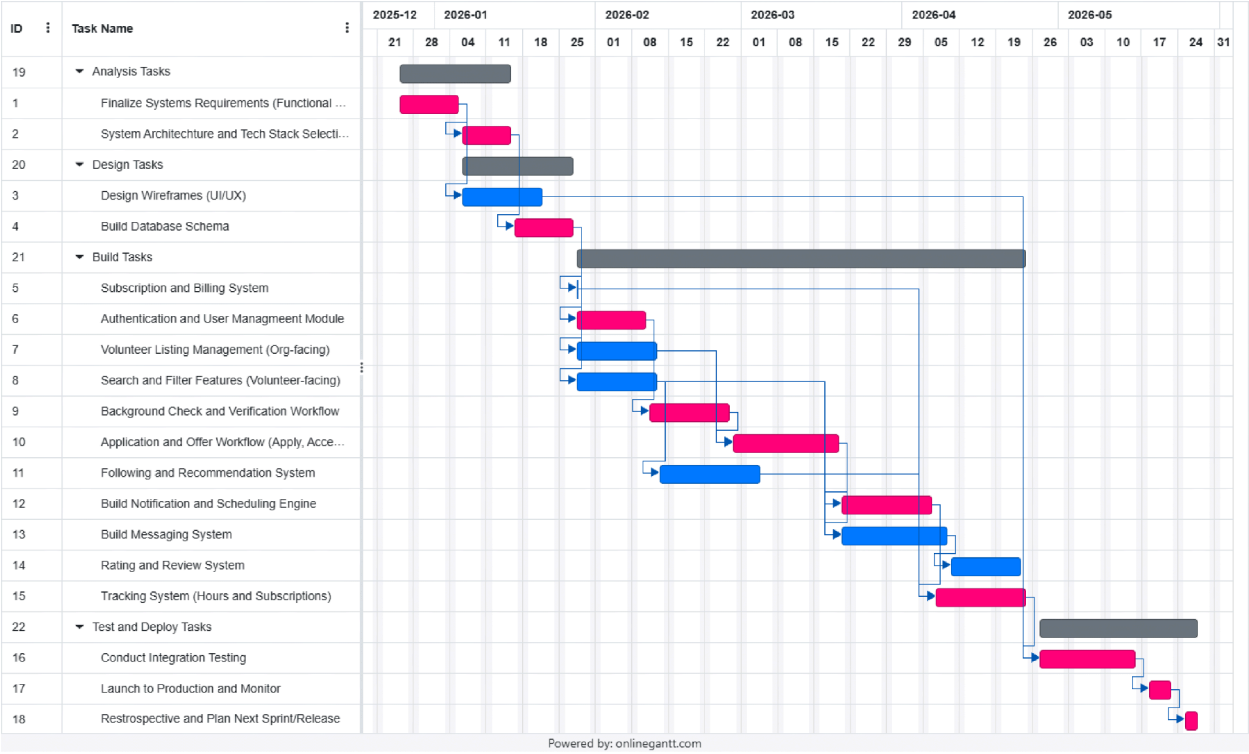
ID	Task Name
A	Requirement Analysis (functional + non-functional)
B	System Architecture & Tech Stack Selection
C	UI/UX Wireframe Design
D	Database Design & Schema Setup
E	Authentication & User Management Module
F	Background Check & Verification Workflow
G	Volunteer Listing Management (Org-facing)
H	Search & Filter Features (Volunteer-facing)
I	Application & Offer Workflow (Apply, Accept, Waitlist)
J	Notification & Scheduling System
K	Messaging System
L	Following & Recommendation System
M	Rating & Review System
N	Subscription & Billing System
O	Tracking System (Hours, Subscriptions, Certifications)
P	Integration & System Testing
Q	Deployment to Production & Monitoring
R	Retrospective & Sprint Planning



Task	Preceding	Time	Te	Tl	Critical?
A	-	10	10	10	YES
B	A	8	18	18	YES
C	A	12	22	89	NO
D	B	8	26	26	YES
E	D	10	36	36	YES
F	E	12	48	48	YES
G	D	12	38	48	NO
H	D	10	36	61	NO
I	F,G	15	63	63	YES
J	I	12	75	75	YES
K	H,I	15	78	99	NO
L	H,I	14	50	75	NO
M	K	10	88	112	NO
N	D	15	41	75	NO
O	N,J,L	14	89	89	YES
P	C,O	15	104	104	YES
Q	P	5	109	109	YES
R	Q	3	112	112	YES

Step 3.5: Build a Gantt chart (via ‘Onlinegantt https://www.onlinegantt.com/#/gantt’ or the tool of your choice) that covers the full SDLC [Google Drive Link for .gantt file]

(open with [https://www.onlinegantt.com/#/gantt])



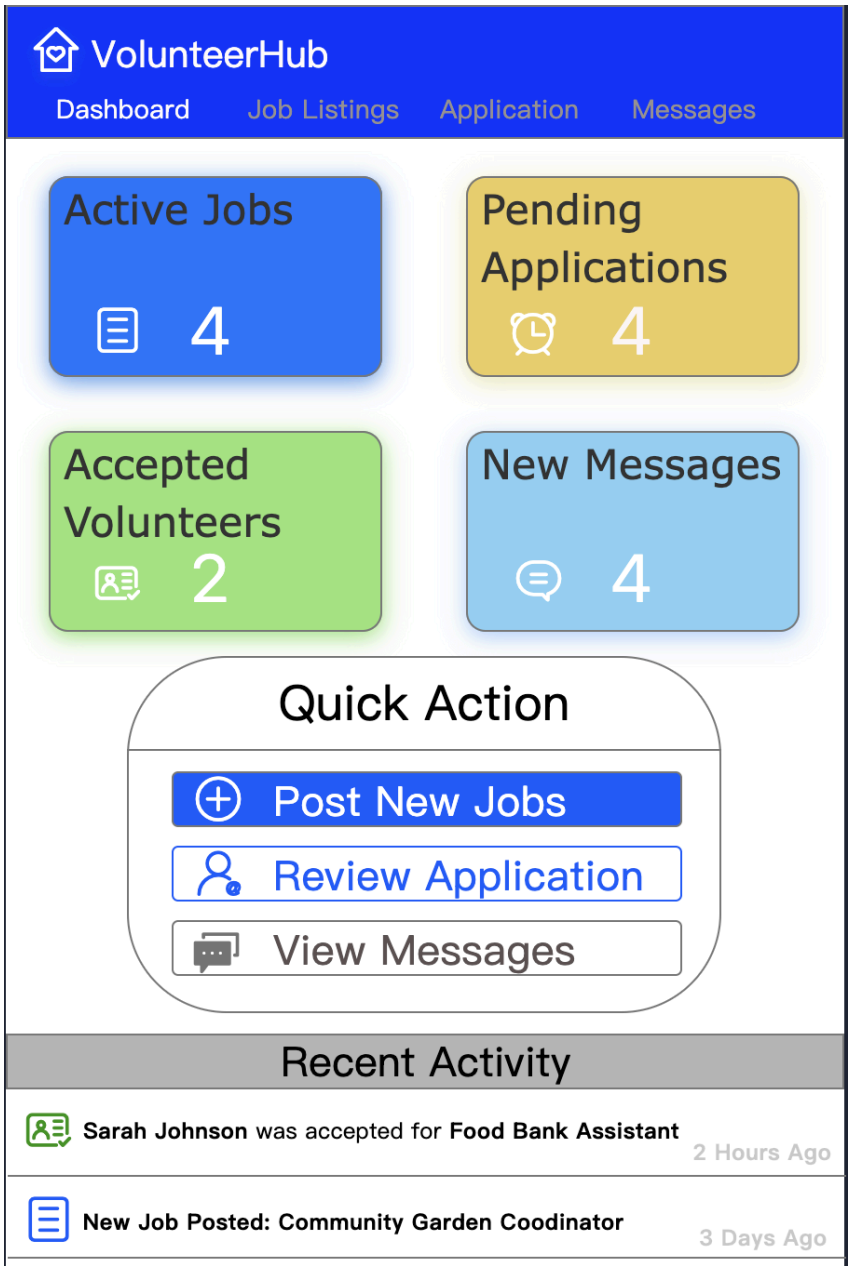
TT1: systems, backend, architecture, infrastructure
TT2: UI/UX, frontend, user-facing interaction layers

Part 4:
[https://rp.mockplus.com/team/invitation/aim-DyvVA/ccqvablbch]

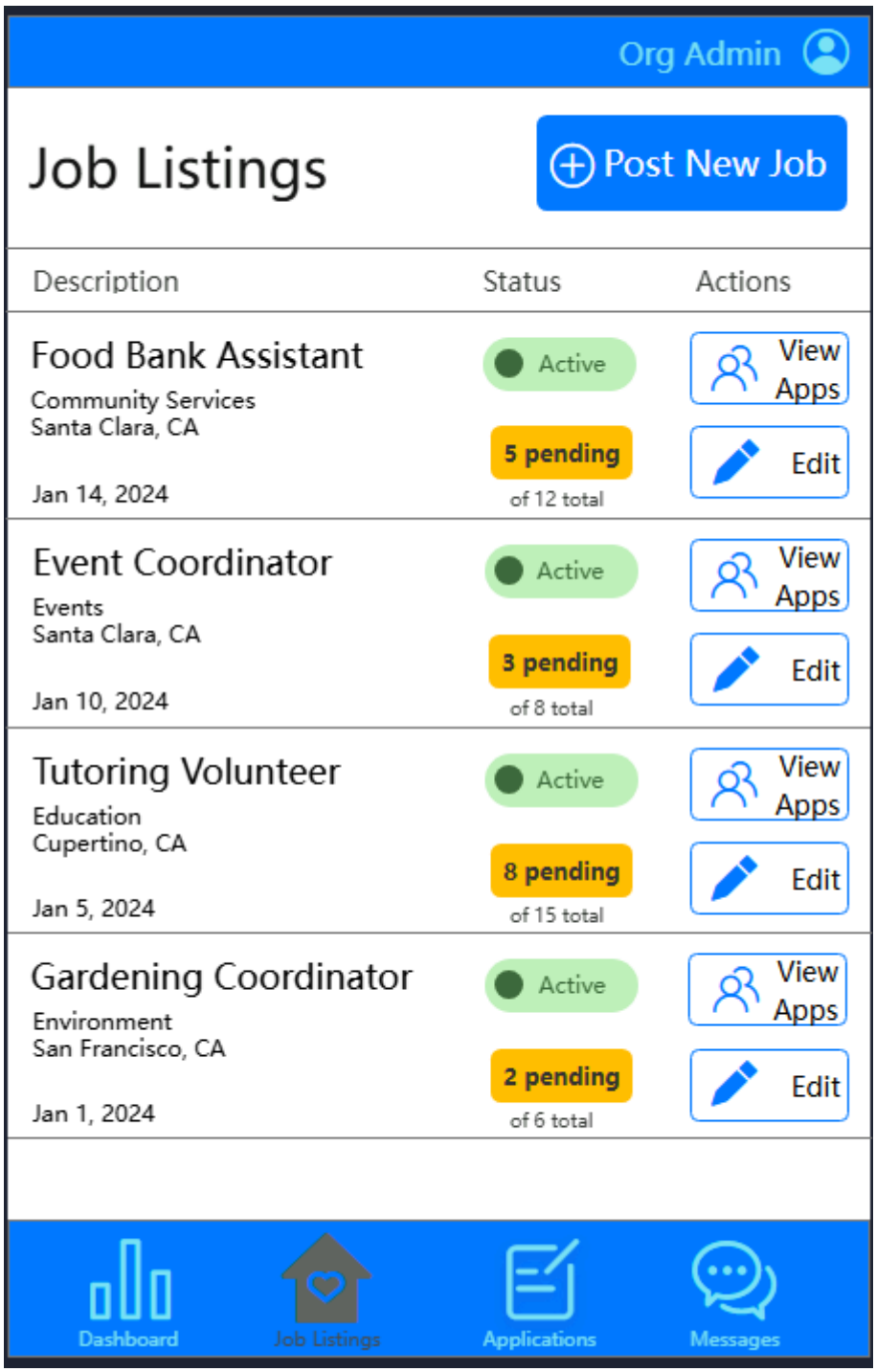
Use the wireframe tool MockPlus ([https://www.mockplus.com/?home=1]) to generate a 3 levels (level-0, level-1, level-2) wireframe pages to show your UI design for your app. In Page Settings, use ‘custom’ to set width = 450, length = 700. Minimal you need to generate 6 pages to illustrate **one (or more) complete use cases**/stories. Use screen capture to capture the proper screen(s), and then import those screenshots into your docx document as the final deliverable.

Use case: Review/Reject/Accept Requests for Volunteer Job

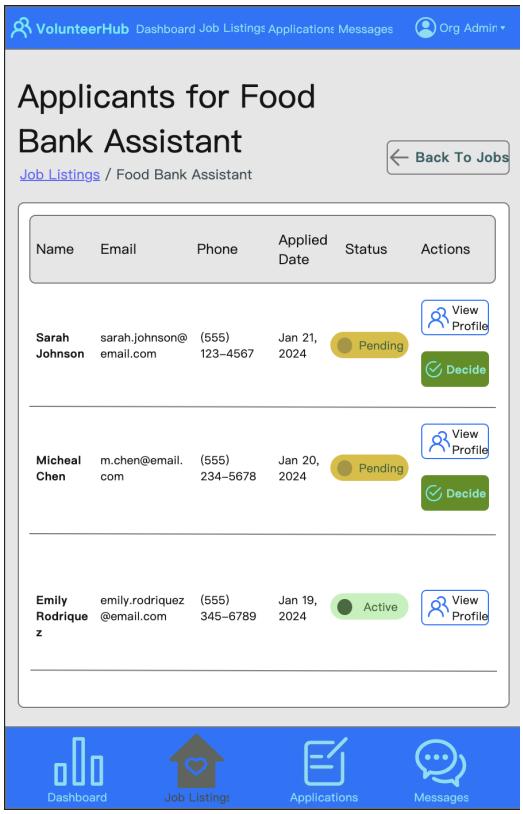
L0: Main org dashboard: navigation to "Job Listings", "Applications", "Messages", etc



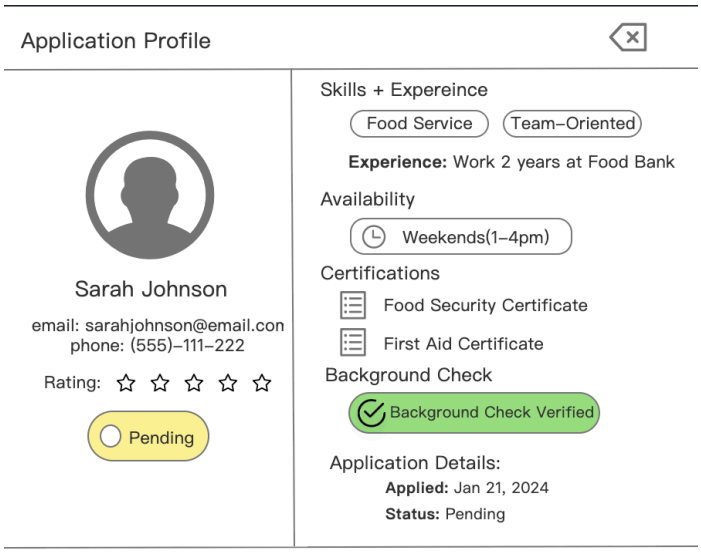
L1: Job Listings List View: Shows posted jobs; includes status and number of pending applications



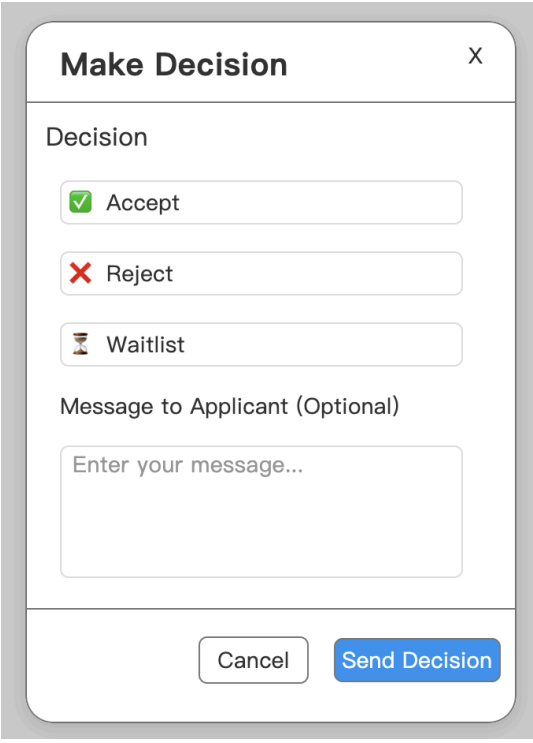
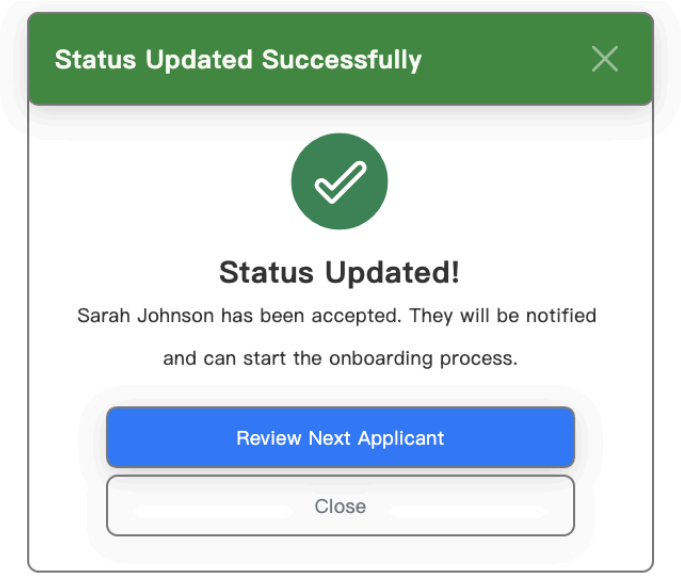
L1: View Applicants for Job: List of volunteers who applied for a selected job



L2: Applicant Profile Preview: Volunteer profile: skills, availability, certs, background check L2 (Chatham)



L2: Make Decision Modal: Admin can choose Accept / Reject / Waitlist + send message L2



L3: Status Updated Confirmation: Shows status update success message and option to review next applicant L3